Detection of Fake Reviews on products using Machine Learning

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Abstract - It's hard to exaggerate how crucial user reviews are in figuring out an organization's e-commerce income. Before purchasing any goods or service, online customers rely on product and service reviews. As a result, firms must consider how reliable internet reviews are because it might directly affect their reputation and bottom line. Because of this, some businesses pay spammers to post false reviews. These false reviews profit from consumer buying choices. As a result, during the past twelve years, there has been substantial study into techniques for spotting false reviews. However, there is still a need for a survey that can analyze and summarize the diverse techniques. In this paper, we're going to put the SVM and Naive Bayesian machine learning model system into place that can spot false reviews with accuracy of 0.801 and 0.687.

Keywords – Reviews, E-commerce, Machine Learning

I. INTRODUCTION

When customers want to make a decision regarding a service or a product, reviews have become the primary source
of information. For example, when consumers take the initiative to book a hotel, they read reviews about previous customers' experiences with the company's services. They select whether or not to reserve a room based on the feedbackfrom the reviews. If they get great feedback from the reviews, they will most likely reserve the room.

As a result, past evaluations have become quite reputablesources of information for most individuals in a variety of internet businesses. Because reviews are regarded real ways of exchanging input about positive or negative services, any attempt to alter such evaluations by providing misleading or inauthentic information is deemed dishonest behaviour, and such reviews are classified as phoney. This situation makes us wonder what if not all of the published reviews are honest or genuine. What if any of these reviews are fictitious? As a result, identifying phoney reviews has become and continuesto be an important and necessary study field.

Machine learning approaches can make a significant contribution to detecting bogus web content evaluations. Web mining approaches, in general, employ a variety of machine learning algorithms to identify and extract importantinformation. Content mining is one of the web mining duties.Opinion mining is a conventional example of content miningsince it is focused with discovering the sentiment of text using machine learning, in which a classifier is trained to examine the characteristics of the reviews as well as the feelings. Detection of phoney reviews is often dependent notonly on the category of reviews but also on some factors thatare not directly related to the content. Text and natural language processing NLP are commonly used in the development of review features.

II. OBJECTIVE AND PROBLEM STATEMENT

Reviews are becoming a more important source of information for consumers. However, reviews that provide afalse impression of a product's quality limit the value of online reviews. Only between 3 and 10 percent of consumers really leave reviews. As of March 2019, there were more thantwo million unconfirmed reviews on Amazon. Therefore, it’s important to find bogus reviews. So to achieve this, we developed a Fake review detector using Machine Learning.

The goal of this paper is to detect the fake reviews. This initiative will transform everyone's lives by making the decisions more effectively. Features Include

- Using Web Interface for review text input with 3different parameters.
- Using Machine Learning techniques to manipulate the text given by the user in our UI.
- Classification of review text as “True” or “Fake”.

III. RELATED WORKS

Unrelated words may appear in a review at times. This gives another consumer the wrong impression, and he orshe may decide not to purchase it. This type of behaviour is commonly referred to as a bogus review. This paper presentsan active learning method for detecting fake and genuine reviews. This include creating a system that only accepts reviews from Authenticated Users. After accepting evaluations from reliable clients (Guanine Users), they will categorize and forecast good, negative, and neutral reviews utilizing NLP- based sentiment analysis and text mining algorithms. Fraudulent, unreliable, and irrelevant word content in fake reviews. They applied only a few fake review detection techniques for amazon dataset. They will try to investigate the possibilities for spotting bogus reviews that are based on actual data by integrating active learning.

This study's findings can advance the science of fake detection on social media platforms [1]. This framework can be utilized for other tasks like the detection of harmful user activity or, as was already noted, the detection of fake news, which may need sophisticated semantic analysis. They suggest expanding the proposed Fake Feature Framework (F3) on a wider domain outside of the e- commerce industry as future work. In this paper, they used four machine learning algorithms to find fake Yelp reviews. The results showed that XGBoost has a very high score in prediction. To train the model SVM algorithm took a long time and Naive Bayes has a low score on average as the dataset is imbalanced. They considered some basic and complex features for training the model, few of them are useful, funny cool rating, similarity of reviewer reviews etc. They had said that we can't predict that the filtered reviews of Yelp are 100 percent fake because there were some factors which lead the Machine learning into false predictions. Taking a proper dataset and applying potentially reliable techniques for filtering review.

In this paper [2], the reviews for a certain product are pulled from the webharvy crawler, coupled with reviews of other products and information about the reviewers to help spot fraudulent reviews. Reviewers using Information Gain and decision tree classifier. The decision's features' significance is verified using information gain. Entropy is calculated for the potential features such as response, template, reply, useful profile etc. to identify whether review is fake or not. The accuracy can be increased by adding more features. The efficiency of the proposed approach had achieved 96% success rate.

One of the primary techniques to tackling the problem has been supervised learning. Obtaining branded false reviews for training [3], on the other hand, is challenging. Because it is extremely difficult, if not impossible, to correctly classify false manual reviews Existing research has employed a variety of instructional purposes, use phony reviews. Possibly the most intriguing kind refers to the phony reviews created by Amazon Mechanical Turk (AMT) is a crowdsourcing platform. Using a custom-made AM claimed an accuracy of 89.6 percent utilizing solely bogus reviews. Consumers may be misled by fake reviews. A high number of bogus reviews might potentially result in massiveproperty losses and public opinion disasters. However,

Because they only employ a single attribute, most existing approaches have a lower accuracy in detecting bogus reviews [4]. A scarcity of labelled experimental data to address this issue, we present a unique approach for detecting reviews. Based on the synthesis of numerous features and rolling collaborative training First, the procedure necessitates the creation of an initial index. System with a variety of characteristics such as text features, emotion features in reviews, and behaviour features reviewers. Second, the procedure necessitates the collection of an initial training sample set. As a result, they created similar algorithms to extract all of a review's characteristics.

Potential customers must base their purchasing decisions on internet reviews [5]. However, its use carries with a curse: misleading opinion spam. Potential customers and companies are misled by opinion restructuring their businesses and guarding against opinion-mining approaches for arriving at correct conclusions as a result, the identification of fraudulent reviews is becoming increasinglyimportant. The purpose of this article was to conduct a linguistic analysis of the nature of fraudulent reviews in the commercial setting of Yelp.com. According to our findings, linguistic characteristics generated a decent 81.3 percent accuracy, which is clearly greater than the 68.1 percent accuracy.

As individuals spend more time shopping and reading reviews online, some reviewers produce fictitious reviews in order to get credit and boost (demote) the sales of products and companies [6]. Detecting fake reviews and spammers becomes more crucial as spamming behavior grows more harmful. Additional work will involve gathering large amounts of review data from various review websites, computer-assisted classification of reviews to minimize the labour of human specialists, and developing a more efficient model for recognizing the association between reviews, reviewers, and retailers.

In this paper, they used three phases of Machine Learning Model for gaining prediction. Steps include Data Pre-processing, using right algorithm like SVM classifier [7], Prediction of Result. They mainly used SVM algorithm because it is used for each classification as well as regression. Subjective prediction is used to pick features for coaching desktop mastering classifiers. As a result, some of the characteristics can also be ineffective for a range of classification systems. For the detection of fraudulent reviews, the cutting-edge find suggests an ensemble pretend overview detection model. Data resampling, characteristic pruning, parameter optimization, and classifier meeting are the techniques in the model. It can improve the text classification as sentiment textual content and make it by means find faux review. They simply acquire the important points only, so we can improvise by taking all points into consideration. They ought to no longer put in force the prediction work, so this can be changed by improvising. Future work may consider the faux evaluate detection that is designed for filtering the pretend reviews. In this SVM classification furnished a higher accuracy of classifying than the Naïve Bayes classifier for trying out dataset. But we can use another different algorithm than these two for higher accuracy.

This paper [8] consists of various techniques and methods which include: feature extraction, review clustering, and classifying fake reviews. One of the most useful techniques to recognize spamming activity in online review is examining the duplicates of reviews. Clustering Methods, Classification Methods are used. Computer generated fake reviews were used to simulate how real-life fake reviews would work. This is because fake review attackers are always looking for new ways without much human involvement. These reviews are generated using language models such as ULMFit and GPT- The two algorithms used to detect the fake reviews are support vector machine (SVM) and OpenAI fake detection model. The dataset used in the experiment, namely the Amazon Review dataset might already contain some fake reviews which can create a bias. There is also a need to develop datasets in various languages as ecommerce is a growing platform in recent times and attracts people from all cultures.

The Amazon Review dataset is used for this experiment [9]. The evaluation approach used in this model is a supervised text approach that classifies the reviews based on the scores given and takes the spam class into consideration.A 10 fold cross validation is performed and the model is proven to be effective. Burst pattern detection is used along with the spam detectors to achieve the needed results. The classification accuracy won't be high as the authenticity can't be determined by the content alone. The Amazon China dataset is used for this experiment [10]. The Unsupervised fake review method was used for detection. A Temporal feature is built and processed. An isolation forest algorithm is used to give an outlier score andthus, the outlier sample. The isolation tree is formed by considering both the temporal features and their respective value. The window size is an important consideration in this model as the optimal results are obtained only in a specific interval (15-30 days). If the window size isn't the same, results may vary.

The proposed model takes the products and the reviews to be evaluated as the input and gives the classification results as the output. The classification is doneby bagging, which considers three bags namely the product word composition classifier (a CNN model), TRIGRAMS and BIGRAMS. A new dataset is created by taking the Amazon dataset into consideration. The only consideration made here was the data searched through fake review keywords in the Amazon dataset. This may not be accurate asthere are way more review or reviewer related features to consider. Sample reviews are generated using language models and then evaluated using qualitative assessments. The better language model is used to create a dataset of fake reviews and some manually written ones (by humans). Some classifier algorithms are trained to differentiate between the two. Next,the accuracy of the classification algorithms are tested against the accuracy of ones annotated by crowd workers. The accuracy of this method can be increased by adding other features such as comparison of reviews, guaranteed purchases etc.

IV. PROPOSED SYSTEM ARCHITECTURE

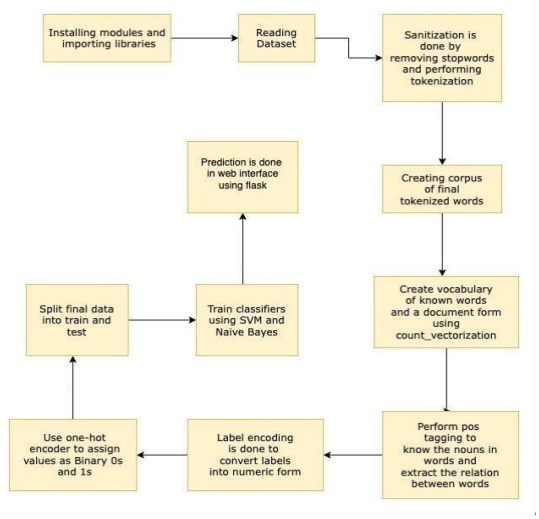

Figure1: Proposed system architecture

Firstly, importing all the libraries and reading the dataset using pandas and removing the unwanted data by removing stop words later tokenizing the sentence by split ting into words and removing it so that a corpus of list of words will be created in figure 1. Now Count Vectorization is used to count number of repeated words and form a document of vocabulary. Perform POS tagging to tag the list of words with their parts of speech and later label encoding happens to convert the labels into machine readable form which are 0s and 1s Once the final data is achieved split the data into train data and test data which is required to train the model using various algorithms like SVM Nave Bayes and now for the prediction and the output we developed a web interface usingpython flask.

1) In this module, we import the necessary libraries for the paper like pandas, math and pickle etc. And we readthe data from the csv file using the pd.read_csv commandand downloading the nltk libraries.

READ DATASET OF CSV FILE USING PANDAS MODULE
THEN:
FIND LEN_DATASET USING MATH FUNCTION.

2) Now, as a part of NLP processing, we will be doing the tokenization which is the process of partitioning text into a sequence of words, whitespaces and punctuation tokens. A tokenization dictionary identifies runs of text that

should be considered words.

FOR I IN RANGE 0 TO LENGTH OF THE DATASET:

REMOVE STOPWORDS FROM THE DATASET REVIEWS

JOINING AND CREATING THE CORPUS OF THE FINAL REVIEW

3) Next, we perform count vectorization which involves counting the number of occurrences of a particular word in a document (i.e. distinct text such as an article, book, even a paragraph!) Python's Sci-kit learn library has a tool called Count Vectorizer to accomplish this.

IMPORT COUNTVECTORIZER
CV=COUNTVECTORIZER(MAX_FEATURES=3000) X=CV.FIT_TRANSFORM(CORPUS).TOARRAY( )

4) Now, we do POS tagging. It is a process of converting a sentence to forms –list of words, list of tuples (where each tuple is having a form (word, tag)). The tag in case of is a part-of-speech tag, and signifies whether the word is a noun, adjective, verb, and so on.

FOR EACH ITEM IN TAGS:
IF EACH ITEM IN VERBS
COUNT VERBS+= 1
ELIF EACH_ITEM IN NOUN
COUNT NOUNS+= 1
ELSE:
CONTINUE

#splitting the dataset as training and testing with 80% fortraining and 20% for testing.

5) Naive Bayes and SVM is used as a classifier for thismodel to predict the outcomes with better accuracy.

Label Encoding refers to converting the labels into a numeric form so as to convert them into the machine- readable form. Machine learning algorithms can then decide in a better way how those labels must be operated.It is an important pre-processing step for the structured dataset in supervised learning.

V. RESULTS AND DISCUSSION

- Programming language used - Python
- The tools used for implementing fake review detection are the Pycharm, Jupyter Notebook.
- Dataset - From Kaggle

Link: https://www.kaggle.com/lievgarcia/amazon-reviews

Implementation

- First, we find the dataset and import the necessary librariesin python to read the data in the dataset and downloading the nltk libraries.
- We will be doing the tokenization i.e., nothing but is the process of partitioning text into a sequence of word, whitespace, and punctuation tokens. A tokenization dictionary identifies runs of text that should be considered words.
- We do count vectorization i.e., counting the number of times each word appears in a document.
- We do the pos tagging. It is a process of converting a sentence to forms – list of words, list of tuples (where eachtuple is having a form (word, tag)). The tag in case of is a part-of-speech tag, and signifies whether the word is a noun, adjective, verb, and so on.
- Label Encoding refers to converting the labels into a numeric form so as to convert them into the machine-readable form. Machine learning algorithms can then decide in a better way how those labels must be operated.
- In one hot encoding, every word (even symbols) which arepart of the given text data are written in the form of vectors,constituting only 1 and 0. So one hot vector is a vector whose elements are only 1 and 0. After one hot encoding divide the training set and test split and train the model.
- Here, we have trained the model and used it for prediction.

After getting the models, we test the trained models using deploy.py. We need a user interface for inputs, so we are using the html and css to create a static web page and using flask we host that page and we will be writing in server.py inwhich we will be using models to predict and return to the page. This Web interface in figure 2 will run on a local host which was shown in above fig. after running the server.py code, so we open through the local host and give some random review inputs in figure 3. Attaching the screenshots of the reviews and the output in our web interface.

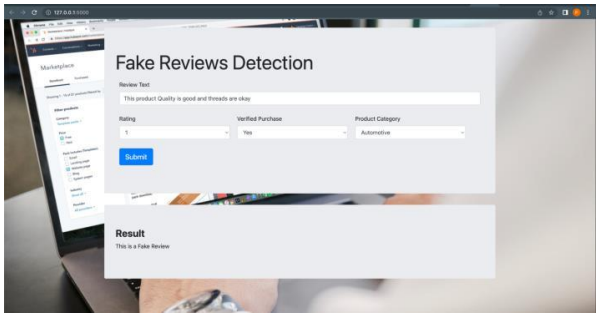

Figure2: Web interface

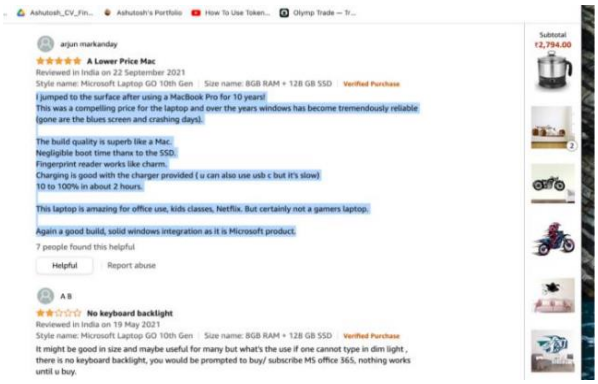

Figure3: Random review inputs

Here, we can see in the below fig that the review is true and it got predicted as true. Because we gave the review text extracted from amazon website in figure 4.

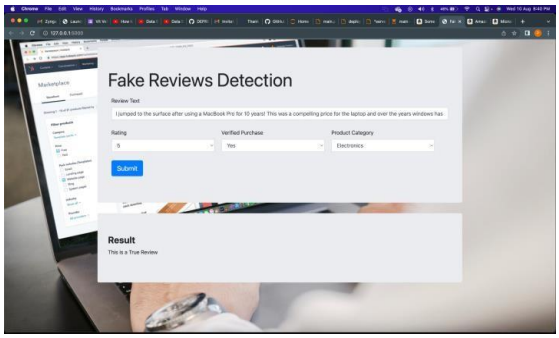

Figure4: Predicted review

Now let's see a review text which was given by us randomly.We give our review as in our own words so that ML model that we trained can detect that it is fake in figure 4 and figure 5.

Figure5: ML model prediction

As you can see the different performance metric values in each algorithm, the SVM classifier gives the best performance out of all the other. The model which is trained by this algorithm is working effectively. Generating the mostaccurate results which we desire for.

CLASSIFIERS	PERFORMANCE METRICS		
	Accuracy	***Recall***	***F-Score***
Naive Byes	0.687	0.722	0.694
SVM	0.801	0.889	0.815

Table1: Performance of various ML model

VI. CONCLUSION

The objective of this paper is to detect the fake review which are present in amazon reviews. This can be achieved by creating a model that can detect the fake reviews by taking input in a text format. The model is trained with numerous reviews, which consists of more than 21000 reviews. In the dataset, the data contains half reviews as fake and one more half as true. So that the model can predict the new data by getting trained with the trained dataset which is amazon reviews dataset. The input review text is taken as sequence of words and perform tokenization and stemming and also removing the stopwords. Later, we do count vectorizer to count the number of repeated words that are present in multiple texts. Perform POS tagging to tag the word with respective parts of speech. Converting into label encoding so that machines can understand the binary format of it after conversion. The training is done with 2 different classifiers like Naïve Bayes and SVM. The fake review detection is intended to screen out fraudulent reviews. For testing the dataset, we employed theSVM classifier in this paper. The SVM classifier, on the other hand, outperformed other algorithms on the training data. Demonstrating that it can generalize better and anticipate fraudulent reviews. This technique can be used onfurther sampled instances of the dataset. Several variables, including the addition of additional vectors such as ratings and verified purchases, have improved the accuracy of data classification.

VII. REFERENCES

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